

Control Light Animations Using Android App

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In this project, an Android application is developed to control light animations using six different LED chasing patterns. Additionally, blinking speed of LEDs is also controlled from the smartphone using fast, moderate and slow buttons on the app. The authors' prototype is shown in Fig. 1.

Circuit and working

The circuit diagram of the project is shown in Fig. 2. Components used in this project are mentioned below.

Arduino Uno. Arduino Uno is an AVR ATmega328P microcontroller (MCU)-based development board with six analogue input pins and fourteen digital I/O pins. The MCU has 32kB ISP flash memory, 2kB RAM and 1kB EEPROM. The board provides serial communication capability via UART, SPI and I2C.

The MCU can operate at a clock frequency of 16MHz. In this project, Arduino receives the command from Android app (in the smartphone) through Bluetooth module, and correspondingly blinks the LEDs (at their specified speed) connected to nine digital pins (5 through 13) of Arduino. Super LEDs can be used to eliminate current-limiting resistors (R1 through R9) in series with them.

PARTS LIST

Semiconductors:

Board1 - Arduino Uno board
LED1-LED9 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):
R1-R9 - 270-ohm

Miscellaneous:

BATT.1 - 9V battery
CON1 - 4-pin connector for Bluetooth module
- HC-05 Bluetooth module
- USB cable
- Breadboard or veroboard
- Jumper wires

(All these are available from kitsnspares.com)

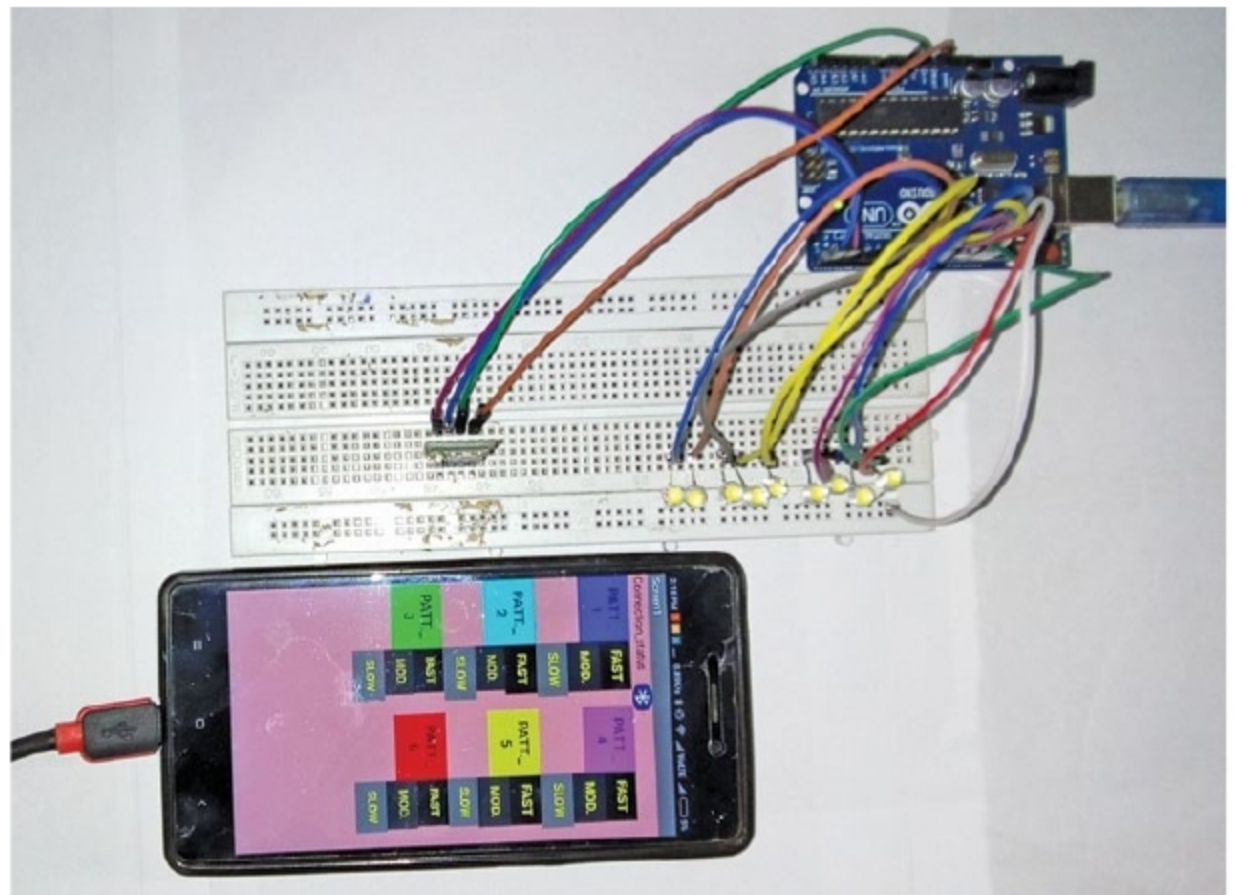


Fig. 1: Authors' prototype wired on breadboard along with smartphone

HC-05 Bluetooth. HC-05 module is an easy-to-use Bluetooth SPP (Serial Port Protocol) module, which is designed for wireless serial connection setup. This module has a fully-qualified Bluetooth V2.0 + EDR (enhanced data rate) 3 Mbps modulation with complete 2.4GHz radio transceiver and baseband. It uses CSR Bluecore 04-External single-chip Bluetooth system with CMOS technology having adaptive

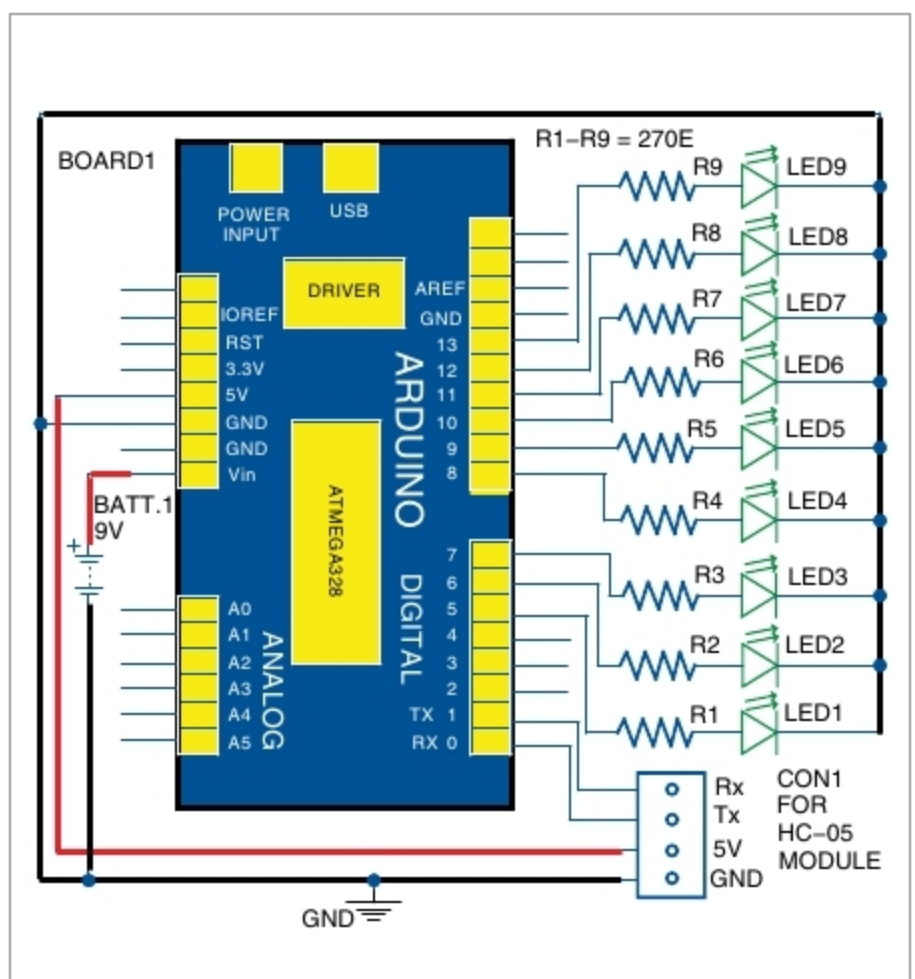


Fig. 2: Circuit diagram of the project